

Connecting Devices

Network Components

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- The network infrastructure contains three categories of hardware components:
 - End devices
 - Intermediate (or Connecting) devices



- Network media



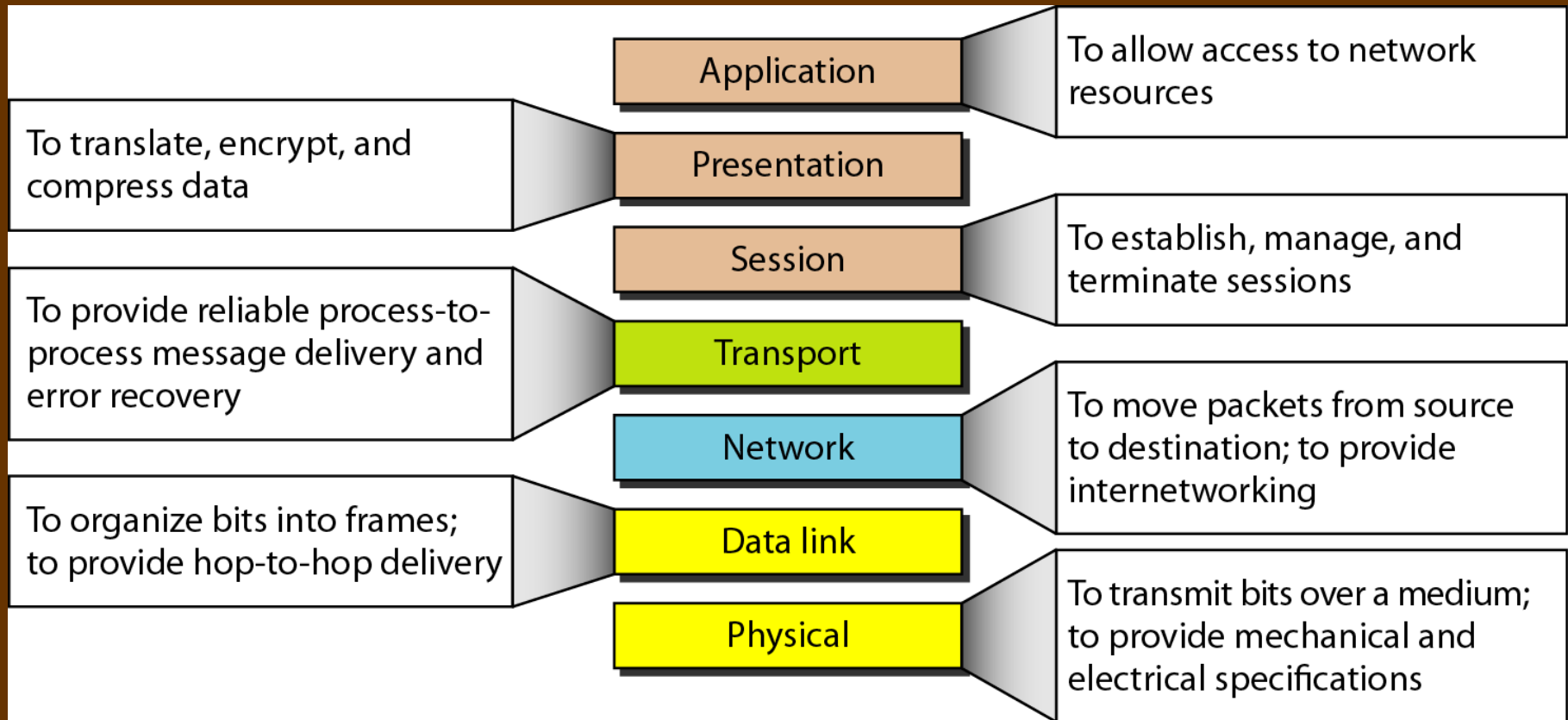
Intermediate (or Connecting) devices

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- **Connecting devices** : are used to connect hosts together to make a network or to connect networks together to make an internetwork(internet).

- Some of the common connecting devices are:
 - Repeaters
 - Hubs
 - Switches
 - Routers
 - Gateways

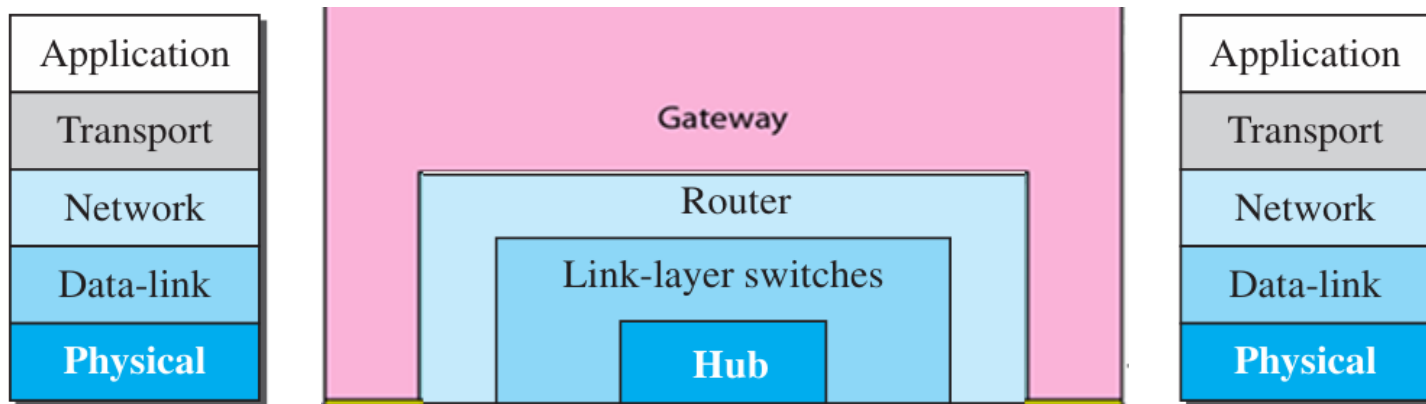
Summary of layers of OSI (Open Systems Interconnection) Model



Intermediate (or Connecting) devices

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Figure 17.1 *Three categories of connecting devices*



Intermediate (or Connecting) devices

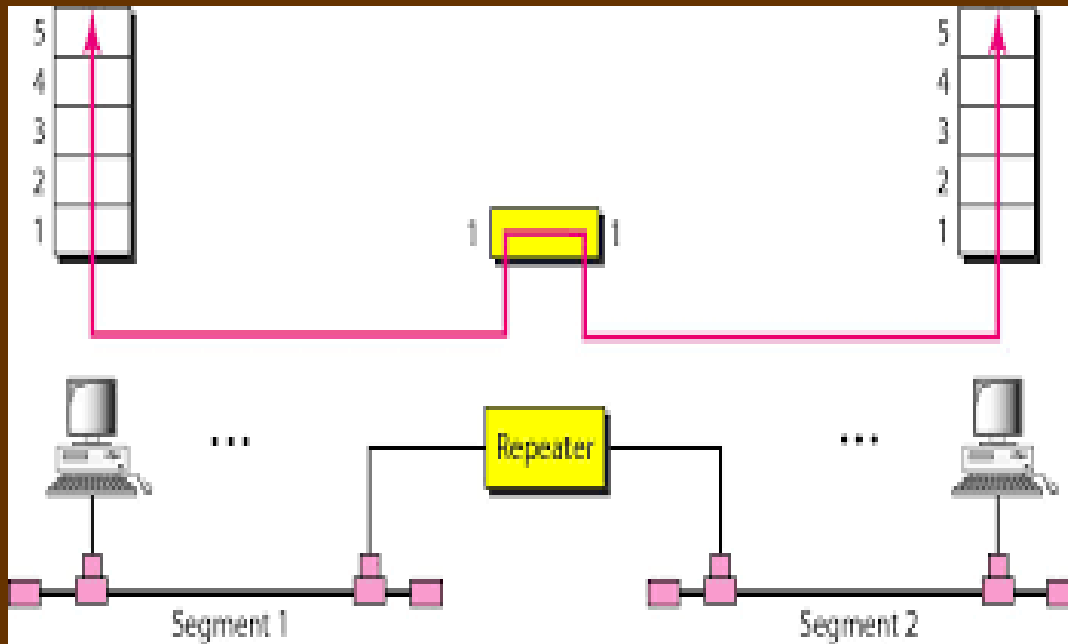
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1. Repeaters

- Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted.
- It helps in extending the length to which the signal can be transmitted over the **same network**.
- An important point to be noted about repeaters is that they do not amplify the signal.
- It is used in Physical Layer

Intermediate (or Connecting) devices

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Repeater Device

Intermediate (or Connecting) devices

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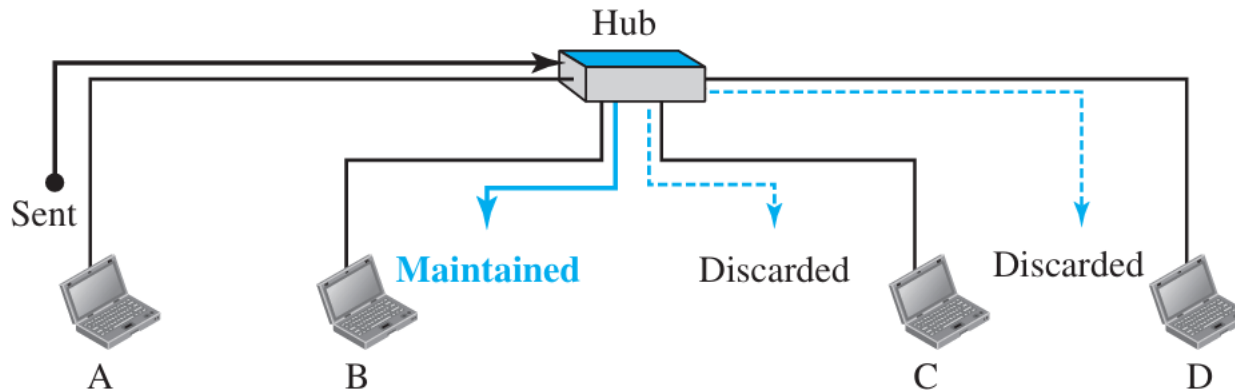
2. Hub

- Like the repeater a **hub** is a device that operates only in the physical layer.
- In the past, when Ethernet LANs were using bus topology, a repeater was used to connect two segments of a LAN to overcome the length restriction of the coaxial cable.
- Today, however, Ethernet LANs use star topology. In a star topology, a repeater is a multiport device, often called a hub, that can be used to serve as the connecting point and at the same time function as a repeater.

Intermediate (or Connecting) devices

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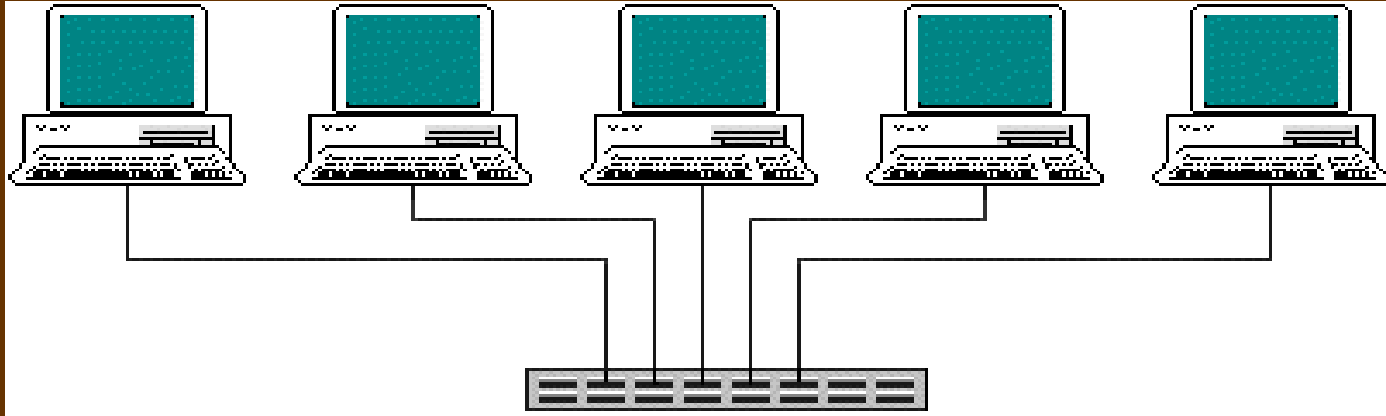
Figure 17.2 *A hub*



- **Station A** sends a packet for **station B** that arrives at the hub. The hub broadcasts the frame. All stations in the LAN receive the frame, but only station B keeps it and others discard.

Intermediate (or Connecting) devices

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Hub Device



Intermediate (or Connecting) devices

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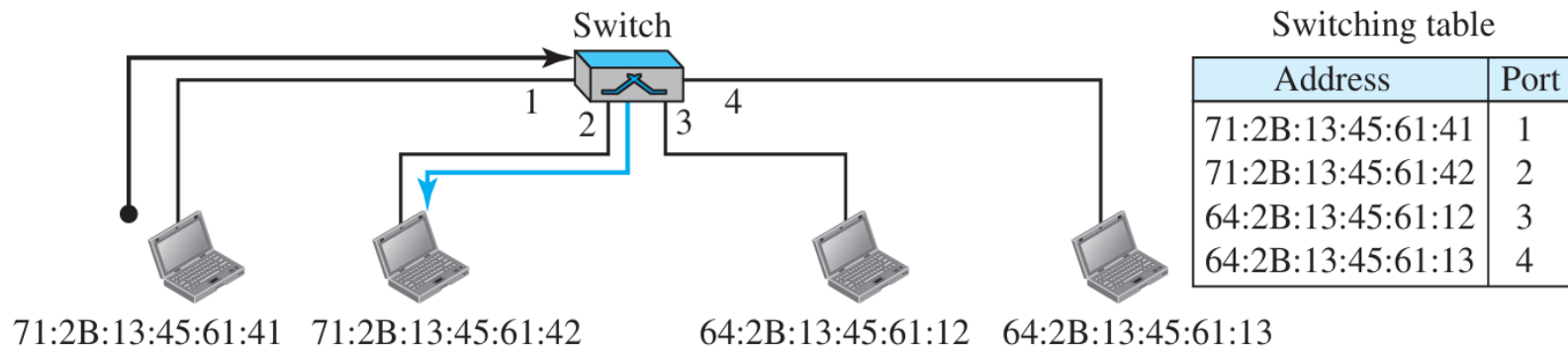
3. Switch

- ❑ Like Hub **Switch** is a multiport device.
- ❑ Switch operates in both the physical and the data-link layers.
- ❑ As a physical-layer device, it regenerates the signal it receives.
- ❑ As a link-layer device, the switch can check the MAC (Media Access Control Identifier) addresses (source and destination) contained in the frame.
- ❑ A switch has filtering capability. It can check the destination address of a frame and can decide from which outgoing port the frame should be sent (next slide).
- ❑ Switches can perform error checking before forwarding data.
- ❑ They are considered very efficient by not forwarding packets that have errors or forwarding good packets selectively to correct devices only.

Intermediate (or Connecting) devices

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Figure 17.3 *Link-layer switch* - Filtering Capability



- If a frame destined for station 71:2B:13:45:61:42 arrives at port 1, the switch consults its table to find the departing port. According to its table, frames for 71:2B:13:45:61:42 should be sent out only through port 2; therefore, there is no need for forwarding the frame through other ports.

Intermediate (or Connecting) devices

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Switch

Intermediate (Connecting) devices

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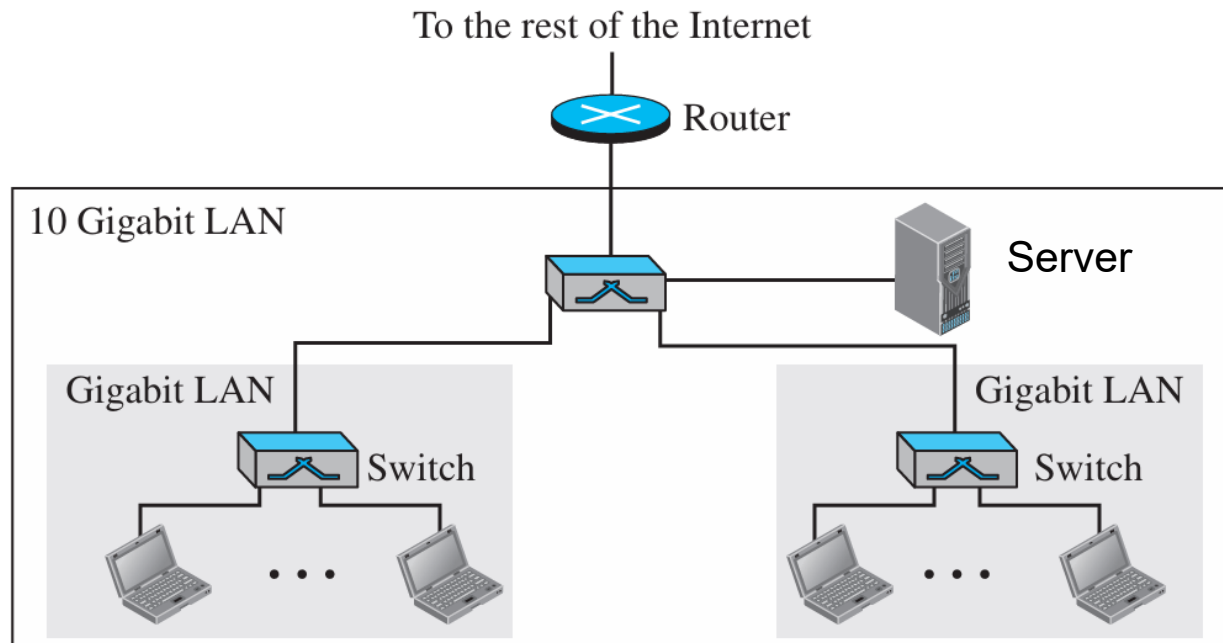
4. Router

- ❑ A **router** is a **three-layer device**; it operates in the physical, data-link, and network layers.
- ❑ As a physical-layer device, it regenerates the signal it receives. As a link-layer device, the router checks the physical addresses (source and destination) contained in the packet. As a network-layer device, a router checks the network-layer addresses (IP addresses).
- ❑ A router is an internetworking device; it connects independent networks to form an internetwork. Routers normally connect LANs and WANs together.
- ❑ They have a dynamically updating routing table based on which they make decisions on routing the data packets.

Intermediate (or Connecting) devices

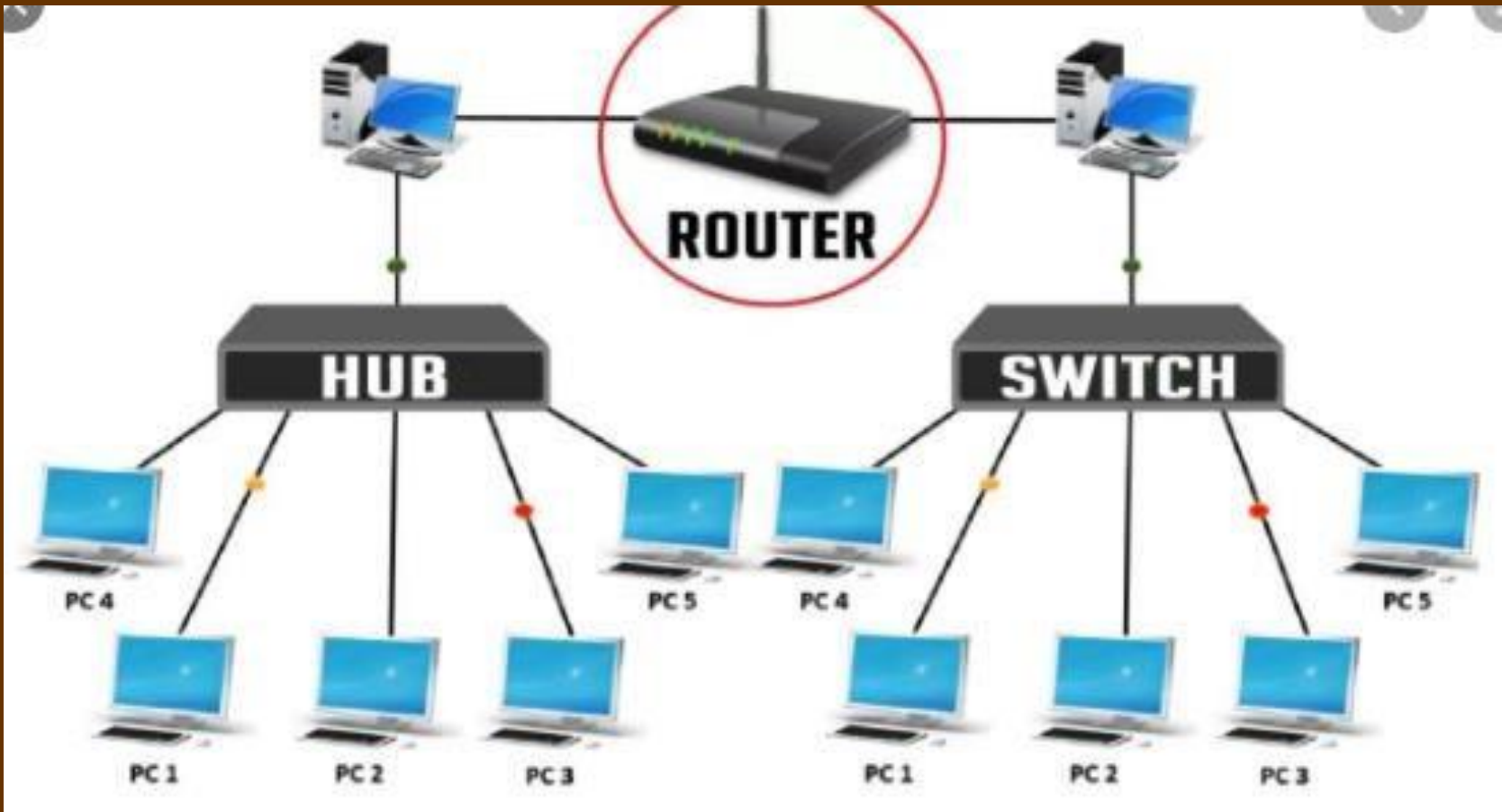
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Figure 17.9 *Routing example*



Intermediate (or Connecting) devices

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Intermediate (or Connecting) devices

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Intermediate (or Connecting) devices

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5. Gateways

- ❑ A Gateway is a single component that enables the data flow from one network to other
- ❑ It act as routers or switches that are capable of interacting with multiple networks that use dissimilar communication protocols.
- ❑ It can work on seven layers of the OSI.
- ❑ The default gateway is a router that connects your host to remote network segments.
- ❑ It can be implemented completely in hardware, software, or as a combination of both.

Connecting devices

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Hub



Gateway



Router



Repeater



Bridge



Switch



Connecting devices

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Repeater	Hub	Switch	Router	Gateway
Physical Layer	Physical Layer	Data Link Layer	Network Layer	ALL 7 layers

